

CoRTeC: Cohort-Conditioned Differentially Private Synthetic Tabular Data from a Frozen Language Model

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Summary. Differentially private synthetic tabular data is worth producing only if a model trained on it performs as one trained on the real records. The deployed marginal mechanisms, MST and AIM, were built for marginal workloads and are evaluated on them; measured instead on this quantity, at library defaults and $\epsilon = 2$ on UCI Adult, their output trains models 0.14 to 0.17 AUC below a same-size real sample. CoRTeC spends the privacy budget once, on a release built for a downstream model, and trains nothing. Within publicly defined cohorts it publishes one Laplace histogram per attribute and outcome class, and a table of target rates over disjoint cells, so conditional structure costs one query per level. A frozen language model then decodes the release under exact per-bin counts and selection from a pool. It never sees a private record, so generation is post-processing: unlimited datasets at no further privacy cost. At $\epsilon = 2$ and $n = 300$, no model trained on CoRTeC's output separates from one trained on a real sample of the same size on census or credit data, one tree model is 0.05 AUC short on clinical data, and 1-way marginal error is within 0.005 of MST's on the first two and below it on the third. The gap lies largely in what is released: giving MST and AIM a private conditional table recovers most of it at no change to either mechanism. Four membership-inference attacks validated on a positive control reach at most 0.049 advantage against a permitted 0.762. Both criteria the field selects synthesisers on are saturated: real data with its target permuted passes a 90% marginal-similarity bar, and a language model given only column names passes aggregate downstream utility on a public benchmark. Only conditional and subgroup measures, against a real-sample and a no-information floor, separate a mechanism that transmits the institution's structure from one that recites a prior.

1. The problem

An institution that holds sensitive tabular data wants to train a model outside its own perimeter, to give a realistic sample to a partner, and to let researchers work on the problem without working on the people. Synthetic data under differential privacy is the accepted way to do all three. Whether it is worth doing depends on one question: if I train on the synthetic data instead of the real data, how much accuracy do I lose?

The mechanisms an institution can deploy today do not answer that question in their own papers. MST and AIM are evaluated on marginal workload error, which they attain, and they were built for published tables rather than for training models. The generative models trained under DP-SGD or PATE are evaluated on downstream utility but, at the budgets and schema sizes we test, do not preserve the target's base rate. The methods that do reach real-data utility fine-tune a language model on the records, which either carries no guarantee or places the private data on the training hardware.

2. What CoRTeC does

CoRTeC separates two things every other method does together: spending the privacy budget and producing records. Stage A spends the budget once, on a release. Within cohorts formed by a public stratification rule it publishes one DP histogram per attribute and per outcome class, a DP class balance and cohort size, and a DP table of target rates over the cells of a conditional hierarchy. Stage B gives a frozen, un-finetuned language model that release and nothing else. Each batch is told the exact number of rows it must produce per bin, per category and per outcome; a threefold pool is filtered to the rows whose cell counts match the release; and values below bin resolution are redrawn inside their released bins. An optional Stage C releases a DP bound on the gap between private and synthetic conditional rates.

Four consequences follow, and they are the reasons to use it.

- *The privacy argument is small.* The generator never sees a private record, so its work is post-processing of a DP release, and the privacy analysis reduces to a composition of counting queries.

- *Generation costs no privacy and can run anywhere.* Any number of datasets can be drawn from one release, and the model can be a tenant-isolated enterprise endpoint or an air-gapped local model with the same guarantee.
- *The budget goes to the structure a classifier needs, and it goes there cheaply.* Histograms have sensitivity 1 regardless of resolution, and disjoint cohorts and cells compose in parallel, so the release carries 6.7× more budget per statistic than the moment-based release it replaced.
- *No statistician is needed.* An auto-configurator chooses the cohorts and the hierarchy from the schema and the budget, charging the one step that reads private data.

3. What we find

Every comparison is at $\epsilon = 2$ and $n = 300$, against a real sample of the same size (the floor no synthetic method should be expected to exceed) and against the same data with its target permuted (the score of data that carries no usable information). Fidelity and utility are always reported together, because a method can score well on one and poorly on the other.

Table 1: 1-way total variation (lower is better) and train-on-synthetic, test-on-real (TSTR) AUC under three students, logistic regression (LR), random forest (RF) and gradient boosting (GBM), higher is better, on the three head-to-head datasets. CoRTeC is the shipped configuration with Gemini 3.5 Flash, three draws per dataset, from releases gated on noised, charged counts and decoded with the exact counts stated as totals over the outcomes. AIM completes on finance only on a 12-column schema, so its finance row is scored on those columns.

dataset	condition	1-way TV	TSTR-LR	TSTR-RF	TSTR-GBM
Adult	real sample	0.043	0.830	0.870	0.840
Adult	CoRTeC	0.026	0.839	0.875	0.844
Adult	MST	0.024	0.690	0.728	0.681
Adult	AIM	0.037	0.675	0.700	0.691
finance	real sample	0.041	0.695	0.727	0.717
finance	CoRTeC	0.029	0.689	0.721	0.699
finance	MST	0.028	0.580	0.610	0.604
finance	AIM (12 columns)	0.054	0.578	0.652	0.623
NHANES	real sample	0.034	0.773	0.728	0.716
NHANES	CoRTeC, auto-configured	0.016	0.738	0.679	0.681
NHANES	MST	0.024	0.687	0.646	0.664
NHANES	AIM	0.024	0.596	0.650	0.650

No student separates from the real-sample floor on Adult or finance; on NHANES the random forest is 0.049 short ($p = 0.033$ before correction) and the other two students within 0.036. Measured on this deliverable at library defaults, models trained on AIM’s and MST’s output score 0.02 to 0.18 AUC lower; on Adult, AIM given a larger model or a 3-way workload and hours to fit reaches the floor too. CoRTeC’s 1-way error is within 0.005 of MST’s on Adult and finance, below it on NHANES, and below a real sample’s on all three. Of 35 Holm-corrected comparisons against AIM and MST on Adult and finance, 29 favour CoRTeC, none favours a baseline, and six do not separate. AIM keeps the lowest error on its own 3-way workloads at adequate sample size, which its design predicts. A powered equivalence test on Adult’s earlier release, six draws against five, gives differences of +0.007, −0.003 and +0.012 AUC with every $p > 0.18$ and every interval inside ± 0.027 .

Four further results reach beyond the mechanism.

- **Both standard acceptance criteria are saturated.** Real data with its target permuted records 95.7% 1-way similarity and passes a 90% bar. On Adult, an ungrounded language model given only the column names reaches TSTR-LR 0.824 against a real sample’s 0.834, AIM’s 0.675 and MST’s 0.690. On a public benchmark, then, neither marginal similarity nor aggregate utility can tell a mechanism that reads the release from one that recites its prior, and that is why the paper’s inversion test exists (Section 6.3): with one released relationship inverted, a matched control with the released arrays deleted emits the inverted relationship at 0.931 where CoRTeC emits 0.000, and only conditional and subgroup measures separate the two.
- **Whether a forced private relationship survives synthesis depends on the mechanism family, not on the guarantee.** MST transmits it with slope 0.998; PATE-CTGAN, at the same ϵ on the same data, with slope 0.054.

What the marginal methods lose is instead a 2.2 to 2.8× inflation of pairwise feature dependence relative to real data; CoRTeC inflates it by 1.24×.

- **Enabling reasoning in the generator changes conditional error 3.8×** (0.173 to 0.045) while downstream AUC cannot detect the change at any feasible draw count.
- **Four membership-inference attacks**, validated on a positive control of real records passed off as synthetic, reach a strongest advantage of 0.049 against a permitted 0.762, with zero exact matches on any dataset.

4. Limitations, in brief

The head-to-head is at $n = 300$ under one generator family; on NHANES three generators (Gemini 3.5 Flash, the open-weight gpt-oss-120b and GPT-5) decode the same release under the per-outcome prompt, no measure separates them after correction at three draws, and the point differences that remain are below what three draws resolve; a second vendor at minimal reasoning reproduces the finance result and the NHANES utility result at three draws each. AIM enters the finance comparison only on a reduced schema and is absent from a fourth, hospital-encounter dataset (Diabetes 130) used in the transmission experiments, because its fit exceeded three hours. ϵ protects a row; on encounter-level data the per-person guarantee is weaker by the maximum contribution count, and the mitigation is a contribution cap taken before the release. Every real dataset is a public benchmark the generator has very likely seen; a constructed registry with no public presence rules out gross recitation but not recall of the marginals. CoRTeC pays per generated record, three times over under the shipped configuration, where the marginal methods pay once and sample freely. The coverage guard that prevents cell-wise generation from silently dropping subpopulations is a heuristic, and when a private conditional table can be conveyed to a language model without silent loss is open. The technical report catalogues twenty-eight measurement artefacts that each produced a plausible but wrong conclusion before it was caught.

5. Artefacts

The paper (43 pages) and the technical report (99 pages, every intermediate experiment and the defect catalogue) are in the research repository, github.com/Calyie/cortec, with the pipeline that produced every table. The two reference implementations, in which every guarantee is enforced in code and every regression test is named after the defect it prevents, are in github.com/Calyie/cortec-framework. Every headline number is re-derived from stored per-draw records by an audit suite, and every number in the paper is checked against the report.